

Engineering Electromagnetics II

Homework #2 Solutions

90 points total

ECE 332 – Spring 2026

Due Tuesday, May 5, 2026

1. Write general expressions for the electric and magnetic fields of a 1 GHz sinusoidal plane wave traveling in the $+y$ direction in a lossless non-magnetic medium with relative permittivity $\epsilon_r = 9$. The electric field is polarized along the x direction, its peak value is 6 V/m, and its intensity is 4 V/m at $t = 0$ and $y = 2$ cm. (10 points)

Here, the angular frequency is

$$\omega = 2\pi f = 2\pi \times 10^9 \text{ rad/s},$$

$\mu_r = 1$, the wavenumber is

$$k = \frac{2\pi}{\lambda} = \frac{2\pi}{\lambda_0} \sqrt{\epsilon_r} = \frac{\omega}{c} \sqrt{\epsilon_r} = \frac{(2\pi \times 10^9 \text{ rad/s})}{(3 \times 10^8 \text{ m/s})} \sqrt{9} = 20\pi \text{ rad/m}$$

and the electric field is

$$\begin{aligned} \mathbf{E}(y, t) &= 6 \cos(\omega t - ky + \phi_0) \hat{\mathbf{x}} \\ &= 6 \cos\left((2\pi \times 10^9 \text{ rad/s})t - (20\pi \text{ rad/m})y + \phi_0\right) \hat{\mathbf{x}} \quad (\text{V/m}). \end{aligned}$$

So, at $t = 0$ and $y = 2 \text{ cm} = 0.02 \text{ m}$,

$$4 \text{ (V/m)} = |\mathbf{E}(0.02 \text{ m}, 0)| = (6 \text{ (V/m)}) \cos(-0.4\pi + \phi_0)$$

So,

$$-0.4\pi + \phi_0 = \cos^{-1}\left(\frac{2}{3}\right) = 0.84 \text{ rad}.$$

This means

$$\phi_0 = 2.1 \text{ rad}$$

and

$$\mathbf{E}(y, t) = 6 \cos \left((2\pi \times 10^9 \text{ rad/s}) t - (20\pi \text{ rad/m}) y + 2.1 \text{ rad} \right) \hat{\mathbf{x}} \text{ (V/m)}.$$

2. The electric field of a plane wave propagating in a lossless, nonmagnetic, dielectric material with $\epsilon_r = 2.56$ is given by

$$\mathbf{E} = 20 \cos \left((6\pi \times 10^9) t - kz \right) \hat{\mathbf{y}} \text{ (V/m)}.$$

(12 points total)

Determine

- (a) f , u_p , λ , k , and η . (5 points)

Here, $\omega = 6\pi \times 10^9 \text{ rad/s}$, $\mu = \mu_0$, and $\mu_r = 1$. So,

$$\begin{aligned} f &= \frac{\omega}{2\pi} = 3 \times 10^9 \text{ s}^{-1}, \\ u_p &= \frac{c}{\sqrt{\epsilon_r}} = \frac{3 \times 10^8 \text{ m/s}}{\sqrt{2.56}} = 1.875 \times 10^8 \text{ m/s}, \\ \lambda &= \frac{u_p}{f} = \frac{1.875 \times 10^8 \text{ m/s}}{3 \times 10^9 \text{ s}^{-1}} = 0.0625 \text{ m} = 6.25 \text{ cm}, \\ k &= \frac{2\pi}{\lambda} = \frac{2\pi \text{ rad}}{0.0625 \text{ m}} = 32\pi \text{ rad/m} = 100.5 \text{ rad/m}, \\ \eta &= \frac{\eta_0}{\sqrt{\epsilon_r}} = \frac{377 \text{ } \Omega}{\sqrt{2.56}} = 235.6 \text{ } \Omega. \end{aligned}$$

- (b) the magnetic field $\mathbf{H}$. (7 points)

In this case,

$$\begin{aligned} \mathbf{H} &= -\frac{20}{\eta} \cos(\omega t - kz) \hat{\mathbf{x}} \text{ (A/m)} \\ &= -\frac{20}{235.6 \text{ } \Omega} \cos \left((6\pi \times 10^9 \text{ rad/s}) t - (32\pi \text{ rad/m}) z \right) \hat{\mathbf{x}} \text{ (A/m)} \\ &= -0.08488 \cos \left((6\pi \times 10^9 \text{ rad/s}) t - (32\pi \text{ rad/m}) z \right) \hat{\mathbf{x}} \text{ (A/m)}. \end{aligned}$$

3. For a wave characterized by the electric field

$$\mathbf{E}(z, t) = a_x \cos(\omega t - kz) \hat{\mathbf{x}} + a_y \cos(\omega t - kz + \delta) \hat{\mathbf{y}},$$

identify the polarization state, determine the polarization angles (γ, χ) , and sketch the locus of $\mathbf{E}(0, t)$ for each of the following cases.

(16 points total)

(a) $a_x = 3 \text{ V/m}$, $a_y = 4 \text{ V/m}$, and $\delta = 0 \text{ radians}$ (4 points)

In this case,

$$\begin{aligned} \mathbf{E}(z, t) &= (3 \cos(\omega t - kz) \hat{\mathbf{x}} + 4 \cos(\omega t - kz) \hat{\mathbf{y}}) \text{ V/m} \\ &= (3\hat{\mathbf{x}} + 4\hat{\mathbf{y}}) \cos(\omega t - kz) \text{ V/m}. \end{aligned}$$

The wave is linearly polarized, the semi-minor axis has length $a_\eta = 0$, the axial ratio is $R = \infty$, and the ellipticity angle χ satisfies

$$\tan \chi = \lim_{R \rightarrow \infty} 1/R = 0 \implies \chi = 0.$$

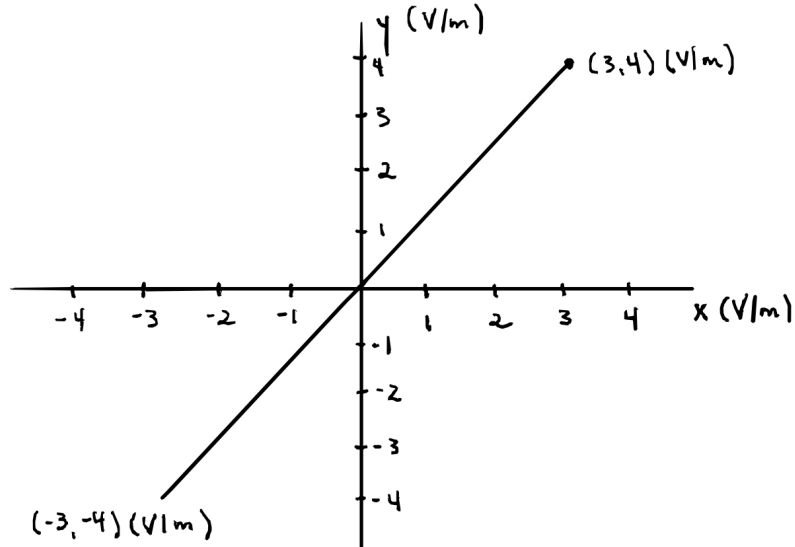
Furthermore, the rotation angle γ is the same as the inclination angle

$$\psi = \tan^{-1}(4/3) \approx 0.9273 \text{ rad} \approx 53.13^\circ.$$

Here,

$$\mathbf{E}(0, t) = (3\hat{\mathbf{x}} + 4\hat{\mathbf{y}}) \cos(\omega t) \text{ V/m},$$

which is the line segment shown below.



- (b) $a_x = 3 \text{ V/m}$, $a_y = 4 \text{ V/m}$, and $\delta = \pi$ radians (4 points)

Here,

$$\begin{aligned}\mathbf{E}(z, t) &= (3 \cos(\omega t - kz) \hat{\mathbf{x}} + 4 \cos(\omega t - kz + \pi) \hat{\mathbf{y}}) \text{ V/m} \\ &= (3 \cos(\omega t - kz) \hat{\mathbf{x}} - 4 \cos(\omega t - kz) \hat{\mathbf{y}}) \text{ V/m} \\ &= (3 \hat{\mathbf{x}} - 4 \hat{\mathbf{y}}) \cos(\omega t - kz) \text{ V/m}.\end{aligned}$$

The wave is linearly polarized, the semi-minor axis has length $a_\eta = 0$, the axial ratio is $R = \infty$, and the ellipticity angle χ satisfies

$$\tan \chi = \lim_{R \rightarrow \infty} 1/R = 0 \implies \chi = 0.$$

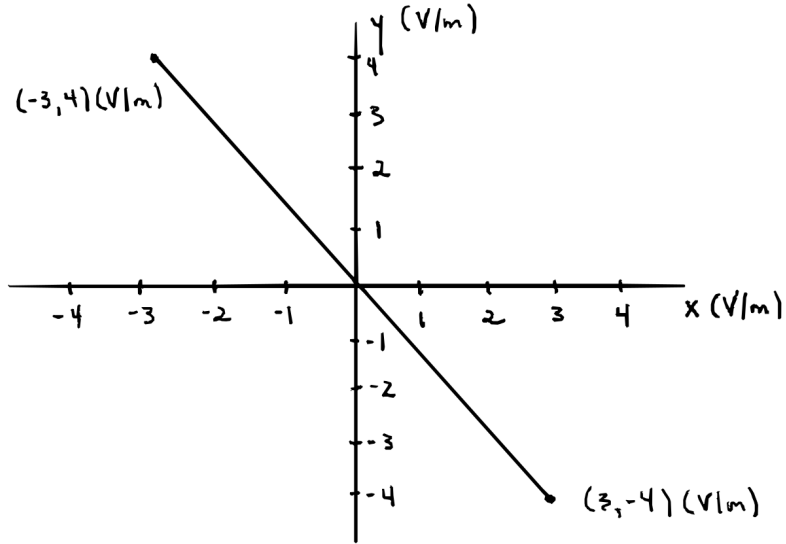
Furthermore, the rotation angle γ is the same as the inclination angle

$$\psi = \tan^{-1}(-4/3) \approx -0.9273 \text{ rad} \approx -53.130^\circ.$$

Here,

$$\mathbf{E}(0, t) = (3 \hat{\mathbf{x}} - 4 \hat{\mathbf{y}}) \cos(\omega t) \text{ V/m},$$

which is the line segment drawn below.



- (c) $a_x = 3 \text{ V/m}$, $a_y = 3 \text{ V/m}$, and $\delta = \pi/4$ radians (4 points)

In this case,

$$\begin{aligned}\mathbf{E}(z, t) &= (3 \cos(\omega t - kz) \hat{\mathbf{x}} + 3 \cos(\omega t - kz + \pi/4) \hat{\mathbf{y}}) \text{ V/m} \\ &= (3 \text{ V/m})(\cos(\omega t - kz) \hat{\mathbf{x}} + \cos(\omega t - kz + \pi/4) \hat{\mathbf{y}}).\end{aligned}$$

The wave is elliptically polarized. The auxiliary angle is

$$\psi_0 = \tan^{-1}(a_y/a_x) = \tan^{-1}(1) = \pi/4 = 45^\circ.$$

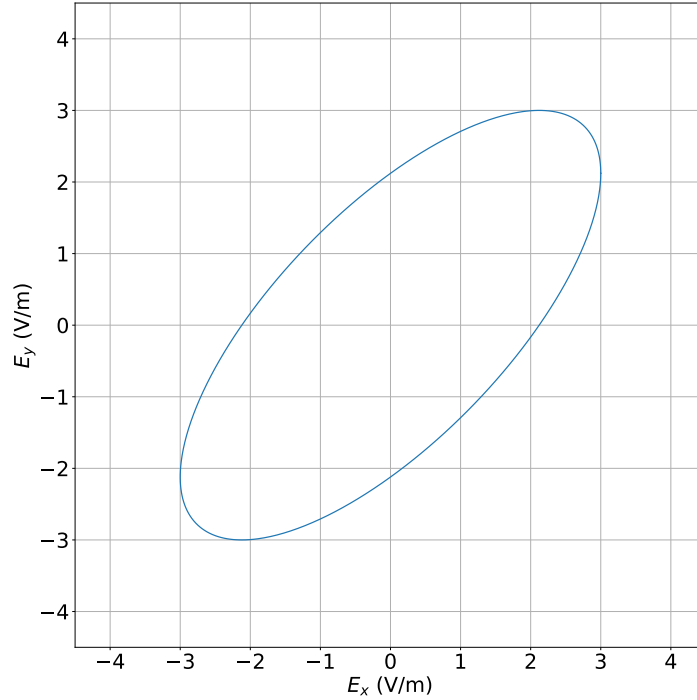
Since $\delta = \pi/4$, $\cos \delta = \sin \delta = \sqrt{2}/2$, $2\psi_0 = \pi/2 = 90^\circ$, and the rotation angle γ and ellipticity angle χ satisfy

$$\begin{aligned}\tan 2\gamma &= (\tan(\pi/2))(\sqrt{2}/2) = \infty \implies 2\gamma = \pi/2 \\ \sin 2\chi &= (\sin(\pi/2))(\sqrt{2}/2) = \sqrt{2}/2 \implies 2\chi = \pi/4.\end{aligned}$$

Therefore, $\gamma = \pi/4$, $\chi = \pi/8$,

$$\mathbf{E}(0, t) = (3 \text{ V/m})(\cos(\omega t) \hat{\mathbf{x}} + \cos(\omega t + \pi/4) \hat{\mathbf{y}}),$$

and the ellipse is the one plotted below.



- (d) $a_x = 3$ V/m, $a_y = 4$ V/m, and $\delta = -3\pi/4$ radians (4 points)

Here,

$$\mathbf{E}(z, t) = (3 \cos(\omega t - kz) \hat{\mathbf{x}} + 4 \cos(\omega t - kz - 3\pi/4) \hat{\mathbf{y}}) \text{ V/m}.$$

The wave is elliptically polarized. The auxiliary angle is

$$\psi_0 = \tan^{-1}(a_y/a_x) = \tan^{-1}(4/3) \approx 0.9273 \text{ rad} \approx 53.130^\circ.$$

Since $\delta = -3\pi/4$, $\cos \delta = \sin \delta = -\sqrt{2}/2$, $2\psi_0 \approx 1.8546 = 106.26^\circ$, and the rotation angle γ and ellipticity angle χ satisfy

$$\tan 2\gamma = (\tan(1.8546))(-\sqrt{2}/2) \approx 2.4244 \implies 2\gamma = -1.1796 \text{ rad} \approx -73.727^\circ$$

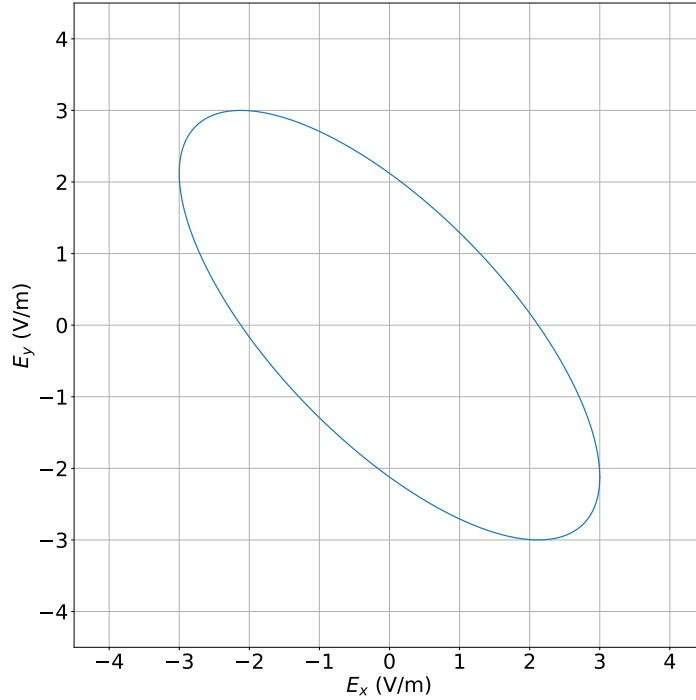
$$\sin 2\chi = (\sin(1.8546))(-\sqrt{2}/2) = -0.74616 \implies 2\chi \approx -0.74616 \text{ rad} \approx -42.752^\circ.$$

Note that the requirement that γ and $\cos \delta$ have the same sign was applied above. Therefore,

$$\gamma \approx -0.3731 \text{ rad} \approx -21.376^\circ, \quad \chi \approx -0.5898 \text{ rad} \approx -36.863^\circ,$$

$$\mathbf{E}(0, t) = (3 \cos(\omega t) \hat{\mathbf{x}} + 4 \cos(\omega t - 3\pi/4) \hat{\mathbf{y}}) \text{ V/m},$$

and the ellipse is the one shown below.



4. A linearly polarized plane wave of the form

$$\tilde{\mathbf{E}} = a_x e^{-jkz} \hat{\mathbf{x}}$$

can be expressed as the sum of an RHC-polarized wave with magnitude a_R and an LHC-polarized wave with magnitude a_L . Prove this statement by finding expressions for a_R and a_L in terms of a_x . (8 points)

Define the RHC wave

$$\tilde{\mathbf{E}}_R = a_R(\hat{\mathbf{x}} + e^{-j\pi/2}\hat{\mathbf{y}})e^{-jkz} = a_R(\hat{\mathbf{x}} - j\hat{\mathbf{y}})e^{-jkz}$$

and the LHC wave

$$\tilde{\mathbf{E}}_L = a_L(\hat{\mathbf{x}} + e^{j\pi/2}\hat{\mathbf{y}})e^{-jkz} = a_L(\hat{\mathbf{x}} + j\hat{\mathbf{y}})e^{-jkz},$$

where

$$a_R = a_L = \frac{a_x}{2}.$$

Then

$$\begin{aligned}\tilde{\mathbf{E}}_R + \tilde{\mathbf{E}}_L &= a_R(\hat{\mathbf{x}} - j\hat{\mathbf{y}})e^{-jkz} + a_L(\hat{\mathbf{x}} + j\hat{\mathbf{y}})e^{-jkz} \\ &= ((a_R + a_L)\hat{\mathbf{x}} + j(-a_R + a_L)\hat{\mathbf{y}})e^{-jkz} \\ &= a_x e^{-jkz} \hat{\mathbf{x}} \\ &= \tilde{\mathbf{E}}.\end{aligned}$$

5. Dry soil is characterized by $\epsilon_r = 2.5$, $\mu_r = 1$, and $\sigma = 10^{-4}$ S/m. At each of the following frequencies, determine if dry soil may be considered a good conductor, a quasi-conductor, or a low-loss dielectric. Then calculate α , β , λ , u_p , and η_c . (24 points total)

- (a) 60 Hz (6 points)
- (b) 1 kHz (6 points)
- (c) 1 MHz (6 points)
- (d) 1 GHz (6 points)

In all cases,

$$\epsilon' = \epsilon = \epsilon_r \epsilon_0 = 2.5 (8.85 \times 10^{-12} \text{ F/m}) = 2.212 \times 10^{-11} \text{ F/m}$$

and

$$\epsilon'' = \frac{\sigma}{\omega} = \frac{\sigma}{2\pi f} = \frac{\frac{1}{2\pi} \times 10^{-4} \text{ S/m}}{f}$$

so that

$$\begin{aligned} \frac{\epsilon''}{\epsilon'} &= \frac{\left(\frac{1}{2\pi} \times 10^{-4} \text{ S/m}\right) / f}{\left(2.212 \times 10^{-11} \text{ F/m}\right)} = \left(\frac{1}{2\pi(2.212)} \times 10^7 \text{ S/F}\right) / f \\ &= (7.193 \times 10^5 \text{ Hz}) / f. \end{aligned}$$

(a) When $f = 60 \text{ Hz}$,

$$\frac{\epsilon''}{\epsilon'} = \frac{7.193 \times 10^5}{60} = 1.199 \times 10^4.$$

So, at this frequency, dry soil is a good conductor. This means

$$\begin{aligned} \alpha &\approx \sqrt{\pi f \mu \sigma} = \sqrt{\pi(60 \text{ s}^{-1})\mu_0(10^{-4} \text{ S/m})} \\ &= \sqrt{\pi(60 \text{ s}^{-1})(4\pi \times 10^{-7} \text{ H/m})(10^{-4} \text{ S/m})} = (\text{Np/m}) \\ \beta &\approx \alpha \approx (\text{rad/m}) \\ \lambda &= 2\pi/\beta = (\text{m}) \\ u_p &= \sqrt{4\pi f / \mu \sigma} = (\text{m/s}) \\ \eta_c &\approx (1 + j) \frac{\alpha}{\sigma} = (\Omega) \end{aligned}$$

(b) When $f = 1 \text{ kHz} = 10^3 \text{ Hz}$,

$$\frac{\epsilon''}{\epsilon'} = \frac{7.193 \times 10^5}{10^3} = 719.3 > 100.$$

So, at this frequency, dry soil is also a good conductor.

(c) When $f = 1 \text{ MHz} = 10^6 \text{ Hz}$,

$$\frac{\epsilon''}{\epsilon'} = \frac{7.193 \times 10^5}{10^6} = 0.7193 \in [10^{-2}, 10^2].$$

So, at this frequency, dry soil is a quasi-conductor.

(d) When $f = 1 \text{ GHz} = 10^9 \text{ Hz}$,

$$\frac{\epsilon''}{\epsilon'} = \frac{7.193 \times 10^5}{10^9} = 7.1993 \times 10^{-4} < 10^{-2}.$$

So, at this frequency, dry soil is a low-loss dielectric.

6. In a nonmagnetic, lossy, dielectric medium, a 300 MHz plane wave is characterized by the magnetic field phasor

$$\tilde{\mathbf{H}} = (\hat{\mathbf{x}} - j4\hat{\mathbf{z}})e^{-2y}e^{-j9y} \text{ (A/m)}$$

Obtain time-domain expressions for the electric and magnetic field vectors. (10 points)

First, the time-domain expression for the magnetic field is

$$\begin{aligned}\mathbf{H}(y, t) &= \text{Re} \left\{ \tilde{\mathbf{H}} e^{j\omega t} \right\} = \text{Re} \left\{ ((\hat{\mathbf{x}} - j4\hat{\mathbf{z}})e^{-2y}e^{-j9y} \text{ (A/m)})e^{j\omega t} \right\} \\ &= \text{Re} \left\{ (\hat{\mathbf{x}} - 4e^{j\pi/2}\hat{\mathbf{z}})e^{j(\omega t - 9y)} \right\} e^{-2y} \text{ (A/m)} \\ &= \text{Re} \left\{ (\hat{\mathbf{x}}e^{j(\omega t - 9y)} - 4e^{j(\omega t - 9y + \pi/2)}\hat{\mathbf{z}}) \right\} e^{-2y} \text{ (A/m)} \\ &= (\cos(\omega t - 9y)\hat{\mathbf{x}} - 4\cos(\omega t - 9y + \pi/2)\hat{\mathbf{z}})e^{-2y} \text{ (A/m)},\end{aligned}$$

where $\omega = 2\pi f = 600\pi \times 10^6 \text{ s}^{-1} = 6\pi \times 10^8 \text{ s}^{-1}$, $\alpha = 2 \text{ Np/m}$, and $\beta = 9 \text{ rad/m}$.

The direction of travel is $\hat{\mathbf{k}} = \hat{\mathbf{y}}$ and

$$\begin{aligned}\tilde{\mathbf{E}} &= -\eta\hat{\mathbf{k}} \times \tilde{\mathbf{H}} = \eta \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ 0 & 1 & 0 \\ e^{-2y}e^{-j9y} & 0 & -4je^{-2y}e^{-j9y} \end{vmatrix} \text{ (V/m)} \\ &= -\eta(-4je^{-2y}e^{-j9y}\hat{\mathbf{x}} - e^{-2y}e^{-j9y}\hat{\mathbf{z}}) \text{ (V/m)} \\ &= \eta(4j\hat{\mathbf{x}} + \hat{\mathbf{z}})e^{-2y}e^{-j9y} \text{ (V/m)}.\end{aligned}$$

The time-domain expression for the electric field is

$$\begin{aligned}\mathbf{E}(y, t) &= \text{Re} \left\{ \tilde{\mathbf{E}} e^{j\omega t} \right\} = \text{Re} \left\{ (\eta(4j\hat{\mathbf{x}} + \hat{\mathbf{z}})e^{-2y}e^{-j9y} \text{ (V/m)})e^{j\omega t} \right\} \\ &= \text{Re} \left\{ \eta(4e^{j\pi/2}\hat{\mathbf{x}} + \hat{\mathbf{z}})e^{j(\omega t - 9y)} \right\} e^{-2y} \text{ (V/m)} \\ &= \text{Re} \left\{ \eta(4e^{j(\omega t - 9y + \pi/2)}\hat{\mathbf{x}} + e^{j(\omega t - 9y)}\hat{\mathbf{z}}) \right\} e^{-2y} \text{ (V/m)}\end{aligned}$$

7. A wave traveling in a nonmagnetic medium with $\epsilon_r = 9$ is characterized by an electric field given by

$$\mathbf{E} = 3 \cos \left((\pi \times 10^7) t + kx \right) \hat{\mathbf{y}} - 2 \cos \left((\pi \times 10^7) t + kx \right) \hat{\mathbf{z}} \text{ (V/m)}.$$

Determine the direction of wave travel and average power density carried by the wave. (10 points)

Since the arguments of the cosine terms contain the spatial terms $+kx$, the direction of travel is the negative x -direction. Also, since there is no attenuation factor, the medium is lossless. The wave is linearly polarized with

$$\begin{aligned}\mathbf{E} &= (3\hat{\mathbf{y}} - 2\hat{\mathbf{z}}) \cos(\omega t + kx) \text{ (V/m)} = \text{Re} \left\{ (3\hat{\mathbf{y}} - 2\hat{\mathbf{z}}) e^{j(\omega t + kx)} \right\} \\ &= \text{Re} \left\{ (3\hat{\mathbf{y}} - 2\hat{\mathbf{z}}) e^{jkx} e^{j\omega t} \right\},\end{aligned}$$

where $\omega = \pi \times 10^7 \text{ rad/s}$. So,

$$\tilde{\mathbf{E}} = (3\hat{\mathbf{y}} - 2\hat{\mathbf{z}}) e^{jkx}.$$

In this case, the intrinsic impedance is

$$\eta = \sqrt{\frac{\mu}{\epsilon}} = \sqrt{\frac{\mu_0}{\epsilon_r \epsilon_0}} = \frac{1}{3} \sqrt{\frac{\mu_0}{\epsilon_0}} = \frac{\eta_0}{3} \approx 40\pi \Omega,$$

$$|\tilde{\mathbf{E}}|^2 = \left| (3\hat{\mathbf{y}} - 2\hat{\mathbf{z}}) e^{jkx} \right|^2 = 9 + 4 = 13,$$

and the average power density of the wave is

$$\mathbf{S}_{av} = \frac{1}{2\eta} |\tilde{\mathbf{E}}|^2 (-\hat{\mathbf{x}}) = -\frac{13}{80\pi} \hat{\mathbf{x}} \text{ W/m}^2 \approx -(51.725 \text{ mW/m}^2) \hat{\mathbf{x}}.$$